



Verbal harassment detection in online games using machine learning methods

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ABSTRACT

Video games have been an inseparable aspect for many throughout their upbringing. The widespread adoption of the internet in the early 2000s has brought video games from the traditional offline media to the online environment. Consequently, people from different parts of the world can play together and communicate in-game with each other. Nowadays, most massively multiplayer online games (MMOs) incorporate voice communication features. Playing video games online with a certain degree of anonymity, along with the ability to verbally communicate with each other, has proven to be a dangerous combination that can breed toxic and abusive behaviors if left unmoderated. This paper proposes a new approach to integrating Whisper, a pre-trained automatic speech recognition (ASR) model, with the well-researched topic of text-based abusive behavior detection. Our proposed verbal harassment detection pipelines yielded an average F-score of 0.899 for all variants tested.

1. Introduction

For many people, playing video games has been an integral part of their upbringing. Video games from traditional offline media have made their way online thanks to the development of the internet [1]. This allows players worldwide to play together and interact with one another in-game [1,2]. The thriving video game industry, supported by the widespread adoption of the internet, had created a new genre of video games called massively multiplayer online (MMO) games. Most MMOs provide the ability to communicate between players, be it through text by means of in-game chat or through audio utilizing in-game voice communication. Unfortunately, recent studies have shown that online-based in-game interaction may pose as a breeding ground for toxic behavior [3,4] such as cyberbullying and online harassment [2,5–7].

Cyberbullying is defined as any act of bullying or targeted harassment that occurs in cyberspace. Several factors might cause someone to commit cyberbullying online against another person, such as improper family education and pedagogical blindness [2]. Furthermore, perpetrators felt a false sense of security whenever they were communicating in-game due to its online nature, triggering a sense of anonymity as players rarely use their own identity in-game. Therefore, in-game communication channels have the unintended effect of providing a social platform that allows players to be as abusive as possible. Online, where things are — due to its associated anonymity — perceived to be

detached from the real world, there are no moral or ethical restrictions, and people openly display their blatant aggression [2].

Numerous studies have been conducted on the task of identifying abusive interactions online [8–15]. However, only a handful of them ever considered audio-based/verbal communication in their studies. Considering that most competitive MMOs rely extensively on voice communication, a new perspective needs to be developed. This study proposes the idea of using a pre-trained automatic speech recognition (ASR) model to overcome the challenges associated with text data gathered from the internet. Avoiding complex data cleaning and pre-processing is one of the strengths of using ASR. The model will output clean dictionary-English words such that problems like character repetition, e.g., “I loveee this!” or the usage of symbols and numbers in place of characters, commonly known as leetspeak, e.g., “Gr8 Succce\$\$!” will not exist in our data.

In this study, we aim to explore the feasibility of integrating the latest pre-trained ASR model from OpenAI, Whisper [16], with natural language processing (NLP) and conventional machine learning (ML) techniques for verbal harassment detection in online games. As the first step, we scrapped audio data from the recorded online games to create a dataset with sufficient profanity to represent the characteristics of online gaming linguistics. Since the study aims to propose an automatic mechanism to detect verbal harassment in a real-world setting, the

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main idea for scrapping this data is to create natural and uncensored data to emphasize the applicability of the proposed approach. As discussed in the methodology section, we employed the dolphin-mixtral model to create an automatic labeling mechanism for in-game utterances. Once the labeled dataset is created in the pre-processing scheme, then we use simple NLP techniques (namely TF-IDF) to extract the features and apply traditional machine learning mechanisms (Support Vector Machines (SVM) and Logistic Regression (LR)) to build a model to distinguish the toxic and non-toxic wordings. The main contributions of this study can be listed as follows:

- A new and uncensored data set on in-game utterances gathered from multiple online sources is presented.
- The efficiency of using the audio data instead of text-based modelings is demonstrated.
- A novel approach for incorporating large language models (LLMs) into an automatic pipeline to label in-game utterances is proposed.
- A comparison of oversampling and undersampling strategies for balancing the offensive and non-offensive data is presented.
- The paper also demonstrates that using traditional ML methods is a computationally efficient way of obtaining high accuracy results which can be useful for real-world scenarios.

The rest of the paper is organized as follows. Related works are reviewed in further detail in Section 2, followed by explanations of the overall methodology in Section 3. Furthermore, in Sections 4 and 5, the results obtained by this study are presented and discussed. In Section 6, a conclusion is provided for this study.

2. Related work

Cyberbullying is a prevalent problem in the online space. Existing works suggest that the underlying cause of the phenomenon of cyberbullying is the environmental factors of the perpetrator [2,4]. But sometimes, in-game factors such as competitiveness [4] may also provoke unruly behavior characterized by abusive communication [2]. A study by Mattinen et al. [3] illustrated the inseparable aspect of verbal communication in modern competitive MMOs. A combination of these factors explains the rise in verbal abuse in an online game environment.

To perform any meaningful operation on text data, a set of features must be extracted. An extensive comparative study by Teng et al. [11] showcased the complexity of feature extractions from text data. Some use statistical methods to extract features from text, such as term frequency and inverse document frequency (TF-IDF) and Bag of Words (BOW) [10,11], and other uses deep learning (DL) methods, such as word to vector (Word2Vec) and Bidirectional Encoder Representations from Transformers (BERT) embedding [12]. Models that use contextual DL embedding, such as BERT, performed ~ 60% better on average relative to those that only use static deep-learned word embeddings, such as Global Vectors for Word Representation (GloVe) and Word2Vec [11]. TF-IDF was utilized because it is less computationally demanding and simpler to implement while maintaining a reasonable performance.

Garbage in, garbage out — a famous adage in the field of Artificial Intelligence (AI) and ML. The quality of the extracted set of features from a text is directly proportional to the quality of the text itself. Current studies [8–15] use manual data preprocessing to handle unnatural word structures. Unnatural word structures are words that contain character repetitions (usually to emphasize something), exhibit the usage of symbols and numbers in place of characters (also known as leetspeak), and include URLs and emojis. This is a sensible approach, considering that most studies were working on text-based data gathered from the internet.

On the other hand, a key advantage of working directly with audio is that those problems mentioned earlier simply do not exist.

Recent advancements in the field of automatic speech recognition by OpenAI [16] have enabled us to reliably retrieve text representation of audio files by means of transcription tasks. This means that, given enough audio samples, we are able to generate text representations to avoid the aforementioned problems.

Existing abusive communication detection methods can generally be classified into two categories: conventional ML and deep learning approaches. Studies that utilize conventional ML classifiers utilize algorithms, such as support vector machines [11] and other related algorithms [9], to make predictions on the input data.

In the study conducted by Reid et al. a logistic regression model was applied to detect toxicity in in-game verbal communication of Overwatch game. With an accuracy performance of 86.3%, their work proved to have a better or at least comparable performance to models in previous studies [17]. Yildiz et al. proposed a novel feature extraction scheme for the classification of audio data collected from YouTube (audio records that included physical violence) as violent and non-violent. The classification accuracies were 89.68% and 89.75% with kNN and SVM approaches respectively [18].

There has been a clear trend of DL-based classifier adoption in recent abusive communication detection studies [8,10,12].

Yousefi et al. proposed a toxic language classifier based on audio data, where the features were extracted directly from audio data. The toxic speech was detected with an accuracy of 75% by utilizing a self-attentive Convolutional Neural Network (CNN) [8]. In another study, Sangeethapriya and Akilandeswari proposed a graph CNN for text classification, pre-trained GoogleNet for image classification, and applied Mel Frequency Cepstral Coefficients and CNN for audio classification in the detection of cyberbullying. The dataset used in this study was collected from various platforms that included Facebook, Twitter, Data Online Harassment, and Cyberbullying detection dataset. Their study yielded an accuracy of 96% [10]. In a similar study, Idrizi and Hamiti utilized text, image, and audio data collected from Facebook, Instagram, and through Google searches. Graph CNN, GoogleNet, and MFCC's were employed where cyberbullying was detected with an accuracy of 83% [19].

Although there has been a noticeable shift towards the adoption of DL-based classifiers, the study from Teng et al. [11] has demonstrated the effectiveness of conventional machine learning algorithms such as SVM and LR. Those two algorithms are considered in this study due to their simplicity and inference speed. Even though the applied approaches (SVM and LR) can be considered more simplistic compared to modern embedding approaches, they provide solid explainability without any extra cost [20]. As discussed in [21] extensively, the underlying mechanisms of LLMs, including the dolphin-mixtral model we have used for labeling purposes, still remain inadequately understood, which raises concerns about potential risks for our implementation area that is strongly related to individuals and their privacy. Consequently, it is imperative to comprehend and elucidate these models to clarify their behavior, limitations, and societal implications. Although several approaches as discussed in [21] are available in the literature for explaining the LLMs, they increase the computational cost. Therefore, we utilized conventional ML approaches, which provide high accuracy along with an explainable mechanism.

As a result, in general terms, the hate speech detection problem is widely studied in the literature [22], but primarily, these studies are based on the text data [23]. There are also several attempts to detect the toxic behavior in audio data, but these methods usually suffer from the regional and cultural differences of the speech [24]. One way to handle this problem is to use the transcription of the audio data [25]. Although the problem discussed in this paper, namely the detection of toxic behaviors in gaming platforms, can be considered a subset of this main problem, there are very few attempts in the literature to adopt these techniques due to the need for high computational speed for real-time systems. In this work, we employed modern ASR techniques (namely the Whisper algorithm) and LLMs (Dolphin-mixtral algorithm) to preprocess the online gaming audio data to handle this problem.

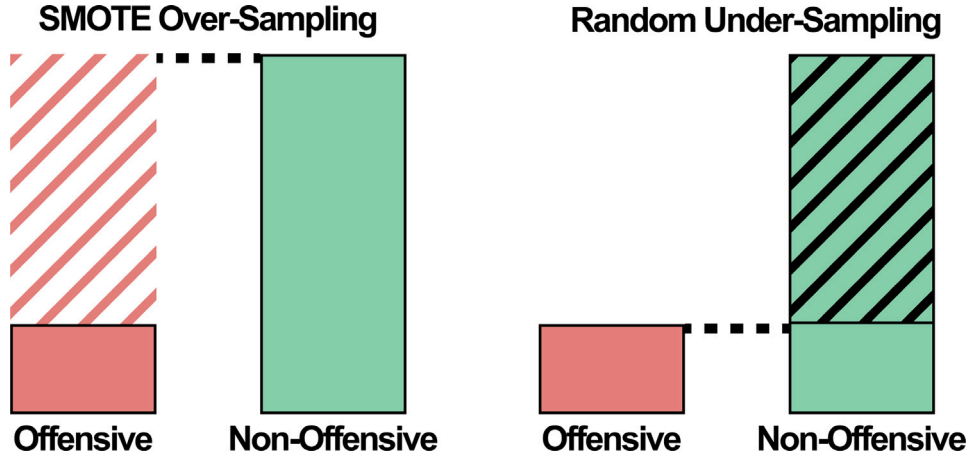


Fig. 1. Visualization of SMOTE and RUS.

3. Methodology

Most public audio datasets lacked sufficient profanity in an in-game communication context. Since it is essential to capture the unique linguistic context of in-game communications, there is a need to develop a dataset that is relevant to this specific style of communication (in-game voice chat). Audio data from more than 250 video recordings of gaming sessions was gathered to generate 126,206 natural in-game utterances. This dataset was collected from multiple online sources using a web-scraping technique.

The audio elements extracted from the recording of the gaming session were processed through a diarization pipeline by utilizing a pre-trained ASR model from OpenAI called Whisper [16]. This model accepts audio as input and will produce a transcription of the audio, complete with the timestamp of each utterance. Although the accuracy of the transcription generated by Whisper was outstanding, their speaker identification algorithm did not meet the expected performance. Hence, Whisper's "best guess" at identifying the speakers was disregarded in this study. Subsequently, text data derived from previous steps were fed into an LLM running the dolphin-mixtral model to estimate their level of toxicity and harassment. The utterances will then be automatically labeled based on the combined consideration of the estimate from the LLM and a list of bad words document provided online by Carnegie Mellon University [26].

From the 126,206 utterances, approximately $\sim 7\%$ of them are labeled toxic, while the rest are labeled non-toxic. This means that our dataset is imbalanced, and to overcome this imbalance, two techniques were considered: under-sampling and oversampling. For this, we utilized Random under-sampling (RUS) and Synthetic Minority Oversampling Technique (SMOTE) approaches [27] (Fig. 1).

In our proposed pipeline (Fig. 2), we assumed that the video game server that handles in-game voice communications could implicitly differentiate speakers by virtue of using different channels for different users. Therefore, there is no need to perform any audio "un-mixing" steps as each audio stream is treated as a unique channel.

The TF-IDF technique was chosen as a feature extraction method for our text inputs due to its simplicity. However, TF-IDF tends to output vector representation with large dimensions depending on the complexity and expansiveness of the training vocabulary. This dimensionality curse is exacerbated even more when we use n-grams to capture possible contexts from a sentence. Thus, dimensionality reduction was considered as a possible solution to this problem, and since TF-IDF naturally outputs a sparse matrix, we implemented the Principal Component Analysis (PCA) via singular value decomposition (SVD) to reduce the computational burden.

All hyperparameters were optimized using a grid search with a 3-fold cross-validation to balance the generalizability of our models and

Table 1

Overview of our data.

	Labels		Count
	Offensive	Not Offensive	
Original	8749	117 457	126 206
Random Undersampling	8749	8749	17 498
SMOTE Oversampling	117457	117 457	234 914

their training time. All models were trained on either undersampled or oversampled data, maintaining a symmetric label ratio of (1:1) for offensive and non-offensive instances, with a holdout rate of 20% (see Table 1).

To summarize, we provided an algorithm given in Alg. 1, which utilizes only the best configuration identified by the proposed pipeline shown in Fig. 2.

Algorithm 1 Proposed approach

Require: Audio data (X)

Ensure: Detection status of verbal harassment (y)

- 1: Convert audio data to text data ($Z \leftarrow \text{whisper}(X)$)
- 2: Label the text data as toxic or non-toxic
 $\hat{y} = \text{DolphinMixtral}(Z)$ $\hat{y} = \{\text{toxic}, \text{non-toxic}\}$
- 3: Extract features from the text data (Z)
 $\hat{Z} = \text{TF-IDF}(Z)$
- 4: Apply oversampling to training and test data formed by $\hat{Z}$ and $\hat{y}$
 $(\hat{Z}_1, \hat{y}_1) = \text{SMOTE}(\hat{Z}, \hat{y})$
- 5: Build the classification model via SVM with 10-fold stratified cross-validation.

4. Results

Our result (Table 2, 3, and 4) shows that the performance of our model using the random under-sampling approach peaked at an F-Score of 0.903 using TF-IDF + SVM. Furthermore, we can also see a reasonably good F-Score of 0.896 by using TF-IDF + PCA + Logistic Regression, considering that the PCA reduces the number of features to only 512. The F-score performance of our model was found to be better in comparison to the result of other existing studies [7–9,14] and inferior to one study [28]. Please refer to Table 5 for further performance comparison.

The performance of SMOTE oversampling method has shown to be better on average compared to that of random under-sampling. However, the result of this approach is problematic and will be discussed further in the discussion section (Section 5).

Table 4 presents the parameters for each of the best-performing models.

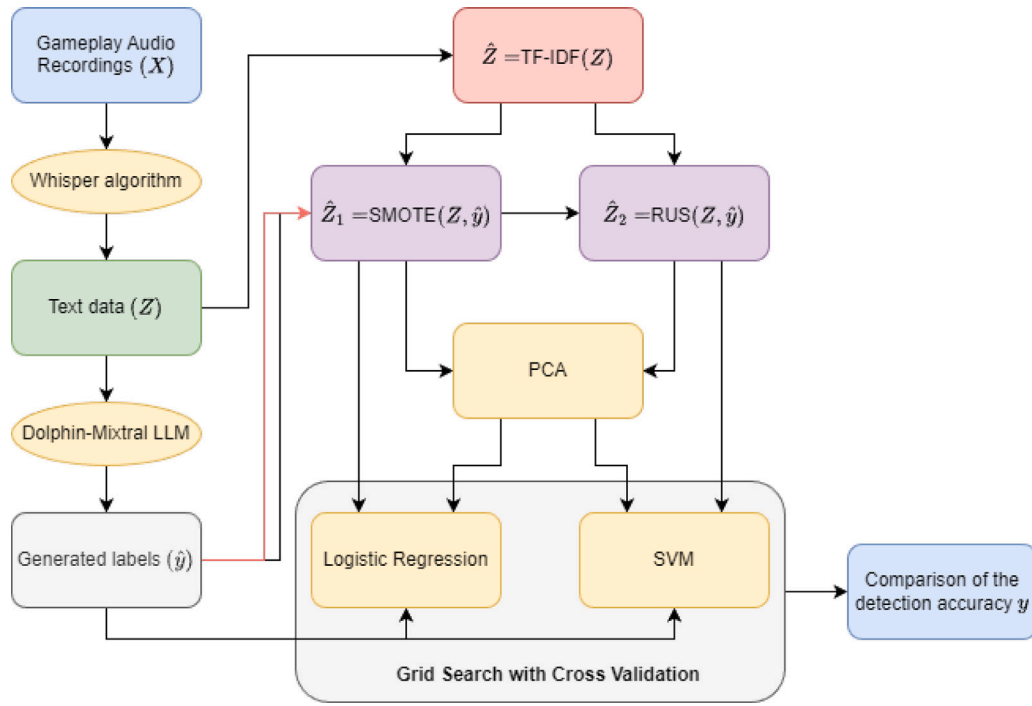


Fig. 2. Proposed pipeline for training.

Table 2

Performance metrics table using under sampling.

	Under Sampling							
	TF-IDF				TF-IDF + PCA			
	Accuracy	Precision	Recall	F1-Score	Accuracy	Precision	Recall	F1-Score
Linear SVM	0.904	0.910	0.896	0.903	0.898	0.923	0.866	0.894
Logistic Regression	0.905	0.917	0.889	0.903	0.899	0.920	0.873	0.896

Table 3

Performance metrics table using over sampling.

	Over Sampling							
	TF-IDF				TF-IDF + PCA			
	Accuracy	Precision	Recall	F1-Score	Accuracy	Precision	Recall	F1-Score
LinearSVM	0.985	0.974	0.997	0.985	0.904	0.936	0.865	0.899
LogisticRegression	0.987	0.978	0.997	0.987	0.907	0.937	0.871	0.903

Table 4

List of parameters for best models according to the grid search.

Undersampling + TF-IDF + SVM:		Undersampling + TF-IDF + Logistic Regression:	
Classifier C	1	Classifier C	10
TF-IDF n-gram	(1,2)	TF-IDF n-gram	(1,1)
Undersampling + TF-IDF + PCA + SVM:		Undersampling + TF-IDF + PCA + Logistic Regression:	
Classifier C	10	Classifier C	1000
TF-IDF n-gram	(1,1)	TF-IDF n-gram	(1,1)
PCA n-components	512	PCA n-components	512
Oversampling + TF-IDF + SVM:		Oversampling + TF-IDF + Logistic Regression:	
Classifier C	100	Classifier C	1000
TF-IDF n-gram	(1,3)	TF-IDF n-gram	(1,3)
Oversampling + TF-IDF + PCA + SVM:		Oversampling + TF-IDF + PCA + Logistic Regression:	
Classifier C	100	Classifier C	100
TF-IDF n-gram	(1,1)	TF-IDF n-gram	(1,1)
PCA n-components	512	PCA n-components	512

5. Discussion

Given the parameters table (Table 4), it is apparent that there is a noticeable discrepancy in parameters between the RUS and SMOTE

sampling strategies. In theory, both RUS and SMOTE should retain the relative structure and characteristics of the data samples. However, as we can see from the table, in the TF-IDF + Logistic Regression model, RUS and SMOTE generated radically different C parameters

Table 5
Relative performance of proposed models.

No	Study	Proposed classifier type	Dataset(s)	% of profanity in dataset(s)	Best F-Score	Relative Performance
1	Smirnov et al. [28]	Deep Learning	The Common Voice, CMU-MOSI	$\pm 25\%$	0.920	+1.9%
2	Proposed approach	Conventional ML	Game Play Recordings	$\pm 7\%$	0.903	0
3	Yildiz et al. [18]	(1)Conventional ML (2)Deep Learning	Youtube	$\pm 38\%$	(1)0.903 (2)0.901	(1)0% (2)-0.2%
4	Pericherla et al. [7]	Deep Learning	Twitter	$\pm 30\%$	0.894	-1%
5	Reid et al. [17]	Conventional ML	Game Play Recordings	50%	0.855	-5.6%
6	Yousefi et al. [8]	Deep Learning	Game Play Recordings	$\pm 81\%$	0.798	-13.2%
7	Rizwan et al. [14]	Conventional ML	(1)Aggression and (2)Twitter Sexism Datasets	Unknown	(1)0.769 (2)0.711	(1)-17.4% (2)-27%
8	Sangeethapriya et al. [10]	Deep Learning	Data Online Harassment (DOH), Cyberbullying Detection Dataset, Social Media, Online Forums	Unknown	0.69	-30.9%
9	Balci et al. [9]	Conventional ML	CCSoft Okey Player Abuse (COPA)	Unknown	0.459	-96.7%

for the same classifier. The discrepancy in other variants was far less noticeable in comparison. The cause of this phenomenon is open for further research.

Furthermore, there are clear indications of an overfit condition with the oversampling models. This is most likely caused by data leakage associated with the oversampling process, whereby there is a possibility of testing data being exposed during the training process due to the inherent “resampling with replacements” that enabled oversampling. Therefore, we decided to disregard the result from our oversampled models and only consider the undersampled models when calculating our relative results (Table 5) and in the conclusion (Section 6).

Although the proposed pipeline has no limitations based on the context of the gaming platform, the main practical issue can be the lower accuracy values of the speech recognition algorithms and LLMs for non-English languages. As widely discussed in the literature, these algorithms (such as Whisper and dolphin-mixtral) are trained with massive corpora of English text and sound data. Therefore they tend to have better accuracy values for English language [29]. This is a well known issue, and in future studies, we are planning to incorporate a language-based metric to assess accuracy in a multilingual set up of our proposed method. For this purpose, text libraries containing slang and offensive words from various languages can be used [30].

The proposed approach has the potential to aid game development companies to better protect their assets, specifically their online game ecosystem from toxic players who can negatively affect the quality of in-game interactions. This would lead to improved player retention and consequently support potential sales for the business. We recommend implementing the proposed system on the game’s communication server, this way, it will essentially act as a moderator capable of detecting toxic behaviors in verbal communication.

Moreover, the impact and result of this study can be further extended by considering more classifier algorithm options such as XG-Boost or Random Forests. It may also be worth exploring deep-learned word embedding models, especially those that are designed to capture word context like the BERT model, to enable a more comprehensive comparison of feature extraction against the currently proposed conventional ML method.

Lastly, the inclusion of end-to-end latency analyses will be beneficial for future studies, considering the latency-sensitive nature of online gaming.

6. Conclusion

This paper proposed a novel pipeline for verbal harassment detection via traditional ML algorithms with efficient integration of several

preprocessing mechanisms. The training data is obtained from recordings of online gaming platforms via web scrapping techniques, and the data is converted to text using the Whisper algorithm. As the second preprocessing operation, the obtained unlabeled text data is automatically labeled with a pre-trained dolphin-mixtral LLM model. Since the obtained data is unbalanced due to the nature of the problem, we compared the performance of SMOTE oversampling and RUS undersampling techniques as the third preprocessing step. Our experiments suggest that even though SMOTE is superior in terms of both classifier accuracy performances, it mainly corresponds to overfitting in the model. Therefore, we used the RUS method to handle the unbalanced structure of the collected data. Moreover, we investigated the effect of dimensionality reduction on accuracy scores. Dimensionality reduction did not lead to any significant improvement or reduction in the performance of undersampled data. However, it can be considered in future studies as a potential approach for reducing the computational cost. The proposed models demonstrated promising results in detecting verbal harassment, achieving the highest F-score of 0.903 and an average F-score of 0.899 across all undersampled variations.

Future studies will focus on addressing the limitations of the study. Although the proposed approach is independent of the context of the gaming platform and the spoken language, as a well-known fact from the literature, both the speech recognition tools and LLMs are trained with massive English texts and sound data sets. Therefore, one can expect their accuracy will not be high enough to detect verbal harassment in non-English languages. As a future study, we plan to develop a language-oriented metric to evaluate the non-English accuracy results. In this study, we have used two well-known and simple classifiers, SVM and LR. Even though the experiments suggest that these two classifiers provide reasonable accuracy scores, in our future studies, we plan to compare these methods with ensembled models and deep-learning-based classifiers, considering factors such as computational burden, accuracy, and explainability.

CRedit authorship contribution statement

Helmi Hibatullah: Validation, Data curation, Writing – original draft, Formal analysis, Writing – review & editing, Investigation, Methodology, Conceptualization. **Tuğçe Ballı:** Supervision, Writing – review & editing, Methodology, Resources, Validation, Formal analysis. **E. Fatih Yetkin:** Validation, Resources, Methodology, Writing – review & editing, Supervision, Formal analysis.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author(s) used OpenAI Whisper and dolphin-mixtral in order to convert audio recordings into texts and then to label said text using a large language model. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

Declaration of competing interest

All authors declare that they have no conflict of interest.

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Data availability

The authors do not have permission to share data.

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